Verification of whether the final matrix is in rref

1. Row 1 and Row 2 are both non-zero rows. ∴ The condition that all non-zero rows must be above the zero rows is vacuously True.

2. Row 1 Leading Entry = 1 in Column 1
   Row 2 Leading Entry = 1 in Column 2
   ∴ The condition that for each row, the leading element must be in a column to the right of the columns of the leading elements of all rows above it is True.

3. Row 2 Column 1 = 0
   ∴ The condition that for each column having a leading entry, all positions below it must contain 0 is True.

4. The condition that each leading entry must be 1 is True.

5. Row 1 Column 2 = 0
   ∴ The condition that for each column having a leading entry, all positions above it must contain 0 is True.

∴ The matrix $\begin{bmatrix} 1 & 0 & 3 & 3 \\ 0 & 1 & 2 & 1 \end{bmatrix}$ is in rref □

By defn, Column 3 has no leading entries. ∴ By defn. x_3 is a free variable.

Since there are no rows in the matrix of the form $[0\ 0 \ldots\ldots 0\ b]$ where $b \neq 0$, ∴ the system is consistent and has infinitely many solutions. (From Theorem 2).

The corresponding system of equations are: $x_1 + 3x_3 = 3$
$x_2 + 2x_3 = 1$

Let, $x_3 = 0$ ∴ $x_2 = 1$, $x_1 = 3$. $x = \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}$. □

Find all x in $\mathbb{R}^4$ that are mapped into the zero vector by the transformation $x \mapsto Ax$ for the given matrix A.

9. $A = \begin{bmatrix} 1 & -4 & 7 & -5 \\ 0 & 1 & -4 & 3 \\ 2 & -6 & 6 & -4 \end{bmatrix}$

Solution: We need to find all such x in $\mathbb{R}^4$ s.t. $Ax = 0$.

By Theorem 3, the matrix equation $Ax=0$ has the same solution set as the system of equations corresponding to the augmented matrix $[A\ 0]$.